

Pygame 시작하기

2025-08-27



Visual Studio Code

Editing evolved

Start

- New File...
- Open File...
- Open Folder...
- Connect to...
- New Workspace with Copilot...

Recent

- ARGallery C:\Users\sunje\UnityProjects
- Pong C:\Users\sunje\UnityProjects
- task0408 C:\Users\sunje\Desktop\SM_unity
- task_ing_after C:\Users\sunje\Desktop\SM_unity
- ShaderTest C:\Users\sunje\UnityProjects
- More...

Walkthroughs

- Get started with VS Code
Customize your editor, learn the basics, and start coding
- Learn the Fundamentals

☒ Show welcome page on startup

- ☰

🔍

🔗

🔧

📁
- New Text FileCtrl+N

New File...Ctrl+Alt+Windows+N

New WindowCtrl+Shift+N

New Window with Profile>

Open File...Ctrl+O

Open Folder...Ctrl+K Ctrl+O

Open Workspace from File...

Open Recent>

Add Folder to Workspace...

Save Workspace As...

Duplicate Workspace

SaveCtrl+S

Save As...Ctrl+Shift+S

Save AllCtrl+K S

Share>

Auto Save

Preferences>

Revert File

Close EditorCtrl+F4

Close WindowAlt+F4

Exit



- Show All CommandsCtrl + Shift + P
- Open FileCtrl + O
- Open FolderCtrl + K Ctrl + O
- Open RecentCtrl + R
- Open ChatCtrl + Alt + I

prac01.py

Create New File (**prac01.py**) Built-In

File 

Editing evolved

 New File... Open File... Open Folder...

>< Connect to...

 Generate New Workspace...

ARGallery C:\Users\sunje\UnityProjects

Pong C:\Users\sunje\UnityProjects

task0408 C:\Users\sunje\Desktop\SM_unity

task_ing_after C:\Users\sunje\Desktop\SM_unity

ShaderTest C:\Users\sunje\UnityProjects

More...

Get started with VS Code

Customize your editor, learn the basics, and start coding

 Learn the Fundamentals

☒ Show welcome page on startup



> 바탕 화면 > Commit



Commit 검색



구성 ▾

새 폴더



홈



갤러리



바탕 화면



다운로드



Documents



사진



이름

"저장"하기 원하는
폴더 위치 지정

수정한 날짜

유형

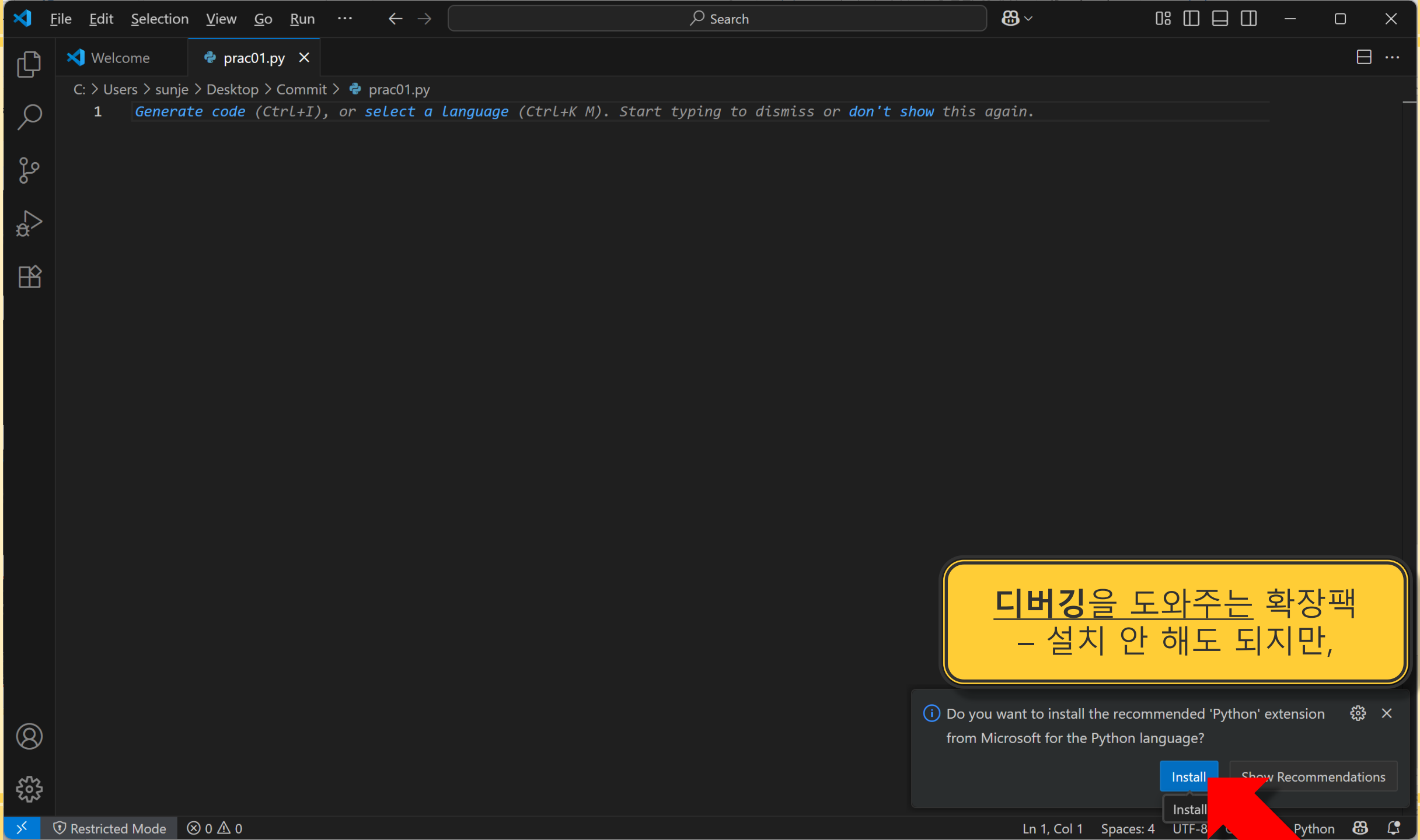
파일 이름(N): prac01.py

파일 형식(I): All Files (*.*)

^ 폴더 숨기기

Create File

취소



디버깅을 도와주는 확장팩
– 설치 안 해도 되지만,

Do you want to install the recommended 'Python' extension from Microsoft for the Python language?

Install Show Recommendations

Install

Install

Restricted Mode

0 0

Ln 1, Col 1 Spaces: 4

UTF-8

Python

설치할 때 화면



Do you trust the authors of the files in this workspace?

Enabling this extension requires a trusted workspace.

If you don't trust the authors of these files, we do not recommend continuing as the files may be malicious. See [our docs](#) to learn more.

Trust Workspace & Install

Install

Learn More

Cancel

Installed extensions

The Python extension will automatically install the following extensions by default to provide the best Python development experience in VS Code:

- [Pylance](#) - to provide performant Python language support
- [Python Debugger](#) - to provide a seamless debug experience with debugpy

These extensions are optional dependencies, meaning the Python extension will remain fully

Marketplace

Identifier	ms-python.python
Version	2025.12.0
Published	2016-01-20, 00:03:11
Last Released	2025-08-21, 19:50:15

Categories

- Programming Languages
- Debuggers
- Data Science
- Machine Learning

Resources

- [Marketplace](#)
- [Issues](#)
- [Repository](#)
- [License](#)

게임 제작 (1)

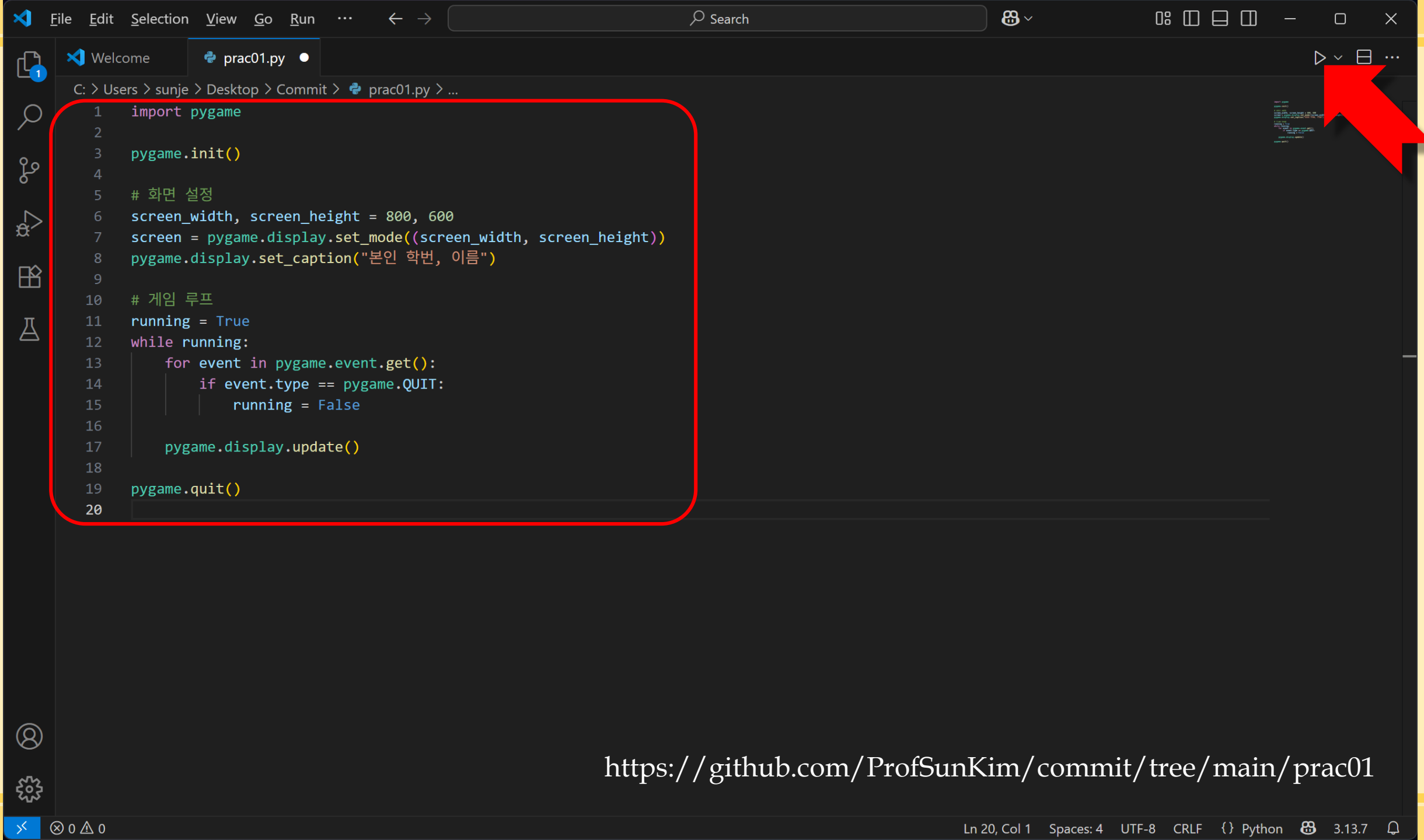
1. 화면 구성 및 캐릭터 이동

- 화면 구성 – 게임 창 설정 후 게임 화면의 크기와 배경색을 설정
- 캐릭터 생성 및 이동 – 간단한 캐릭터를 화면에 표시하고, 키보드 입력을 받아 캐릭터를 상하좌우로 이동

2. 맵 생성 및 충돌 처리

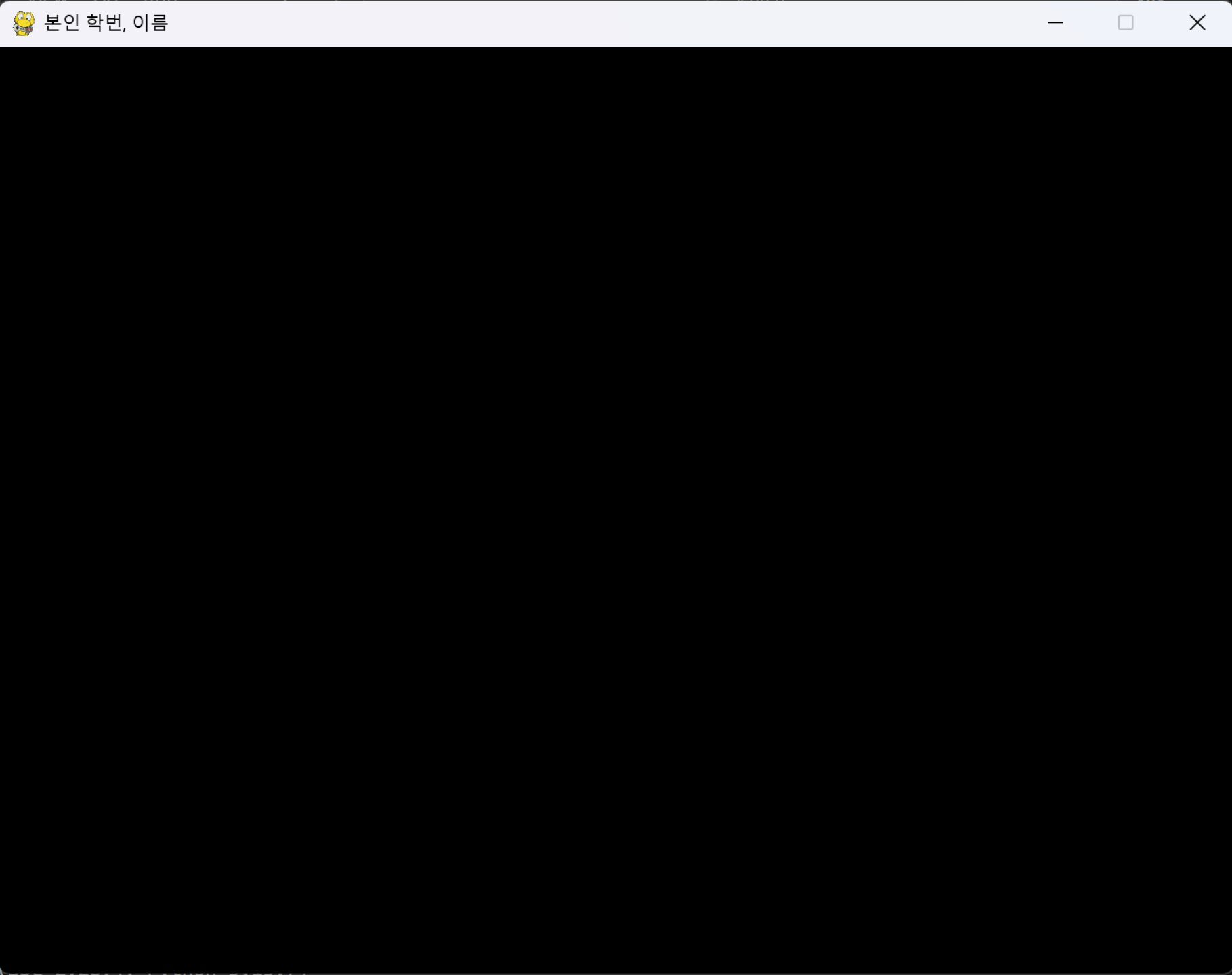
3. 적 생성 및 전투 시스템

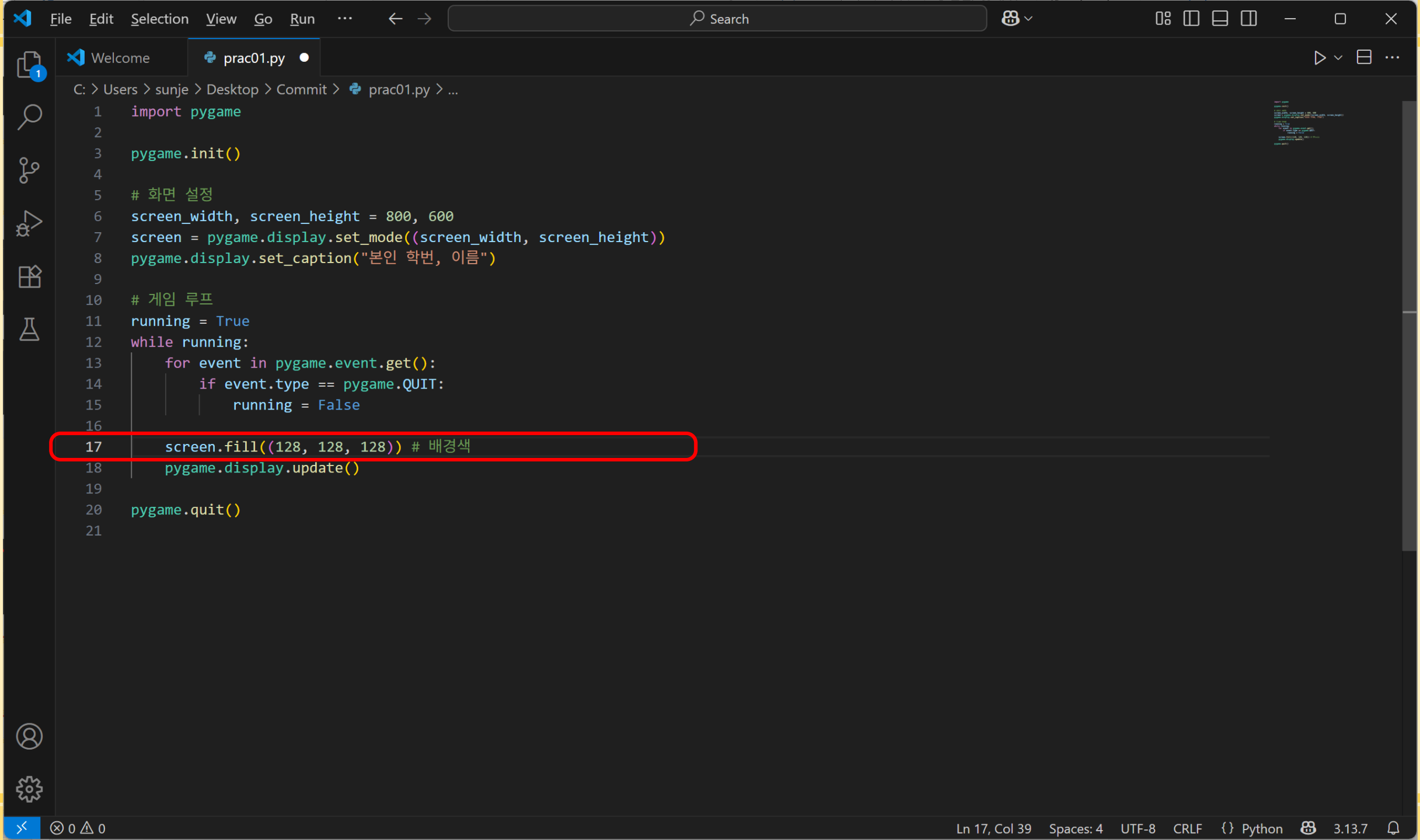
4. 아이템 생성 및 게임 완성



```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9
10 # 게임 루프
11 running = True
12 while running:
13     for event in pygame.event.get():
14         if event.type == pygame.QUIT:
15             running = False
16
17     pygame.display.update()
18
19 pygame.quit()
20
```

<https://github.com/ProfSunKim/commit/tree/main/prac01>

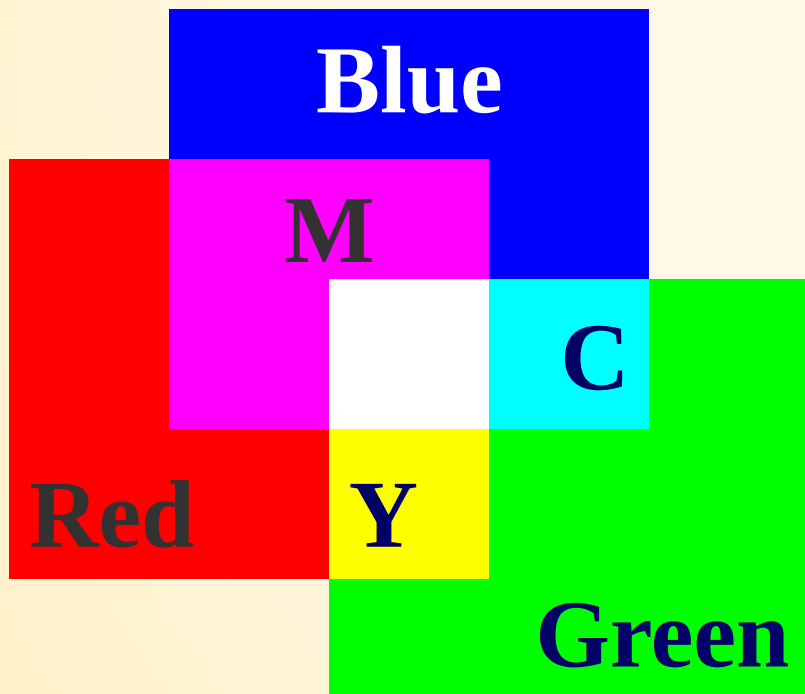






RGB 색상

- Red(빨강), Green(녹색), Blue(파랑)
 - 0~255 사이의 정수



Color	Red	Green	Blue
Red	255	0	0
Green	0	255	0
Blue	0	0	255
Magenta	255	0	255
Cyan	0	255	255
Yellow	255	255	0
White	255	255	255
Black	0	0	0

연습 문제 (1)

- 배경색을 원하는 색으로 변경하기
- 반드시, 자신의 학번과 이름으로 캡션도 함께 변경하기!!!



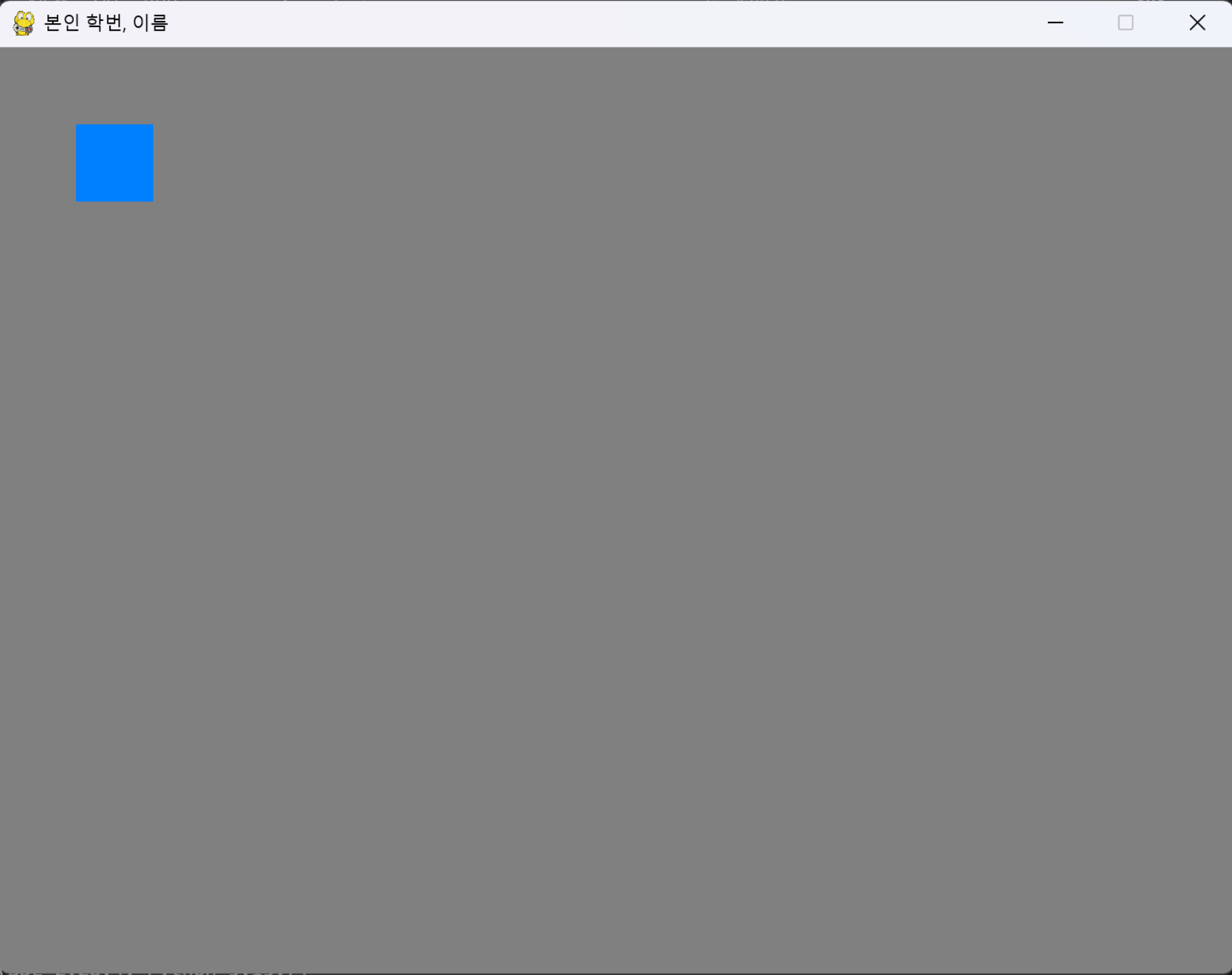
Welcome

prac01.py

prac01-1.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9
10 # 게임 루프
11 running = True
12 while running:
13     for event in pygame.event.get():
14         if event.type == pygame.QUIT:
15             running = False
16
17     screen.fill((128, 128, 128)) # 배경색
18     pygame.draw.rect(screen, (0, 128, 255), (50, 50, 50, 50)) # 캐릭터
19     pygame.display.update()
20
21 pygame.quit()
22
```



연습 문제 (2)

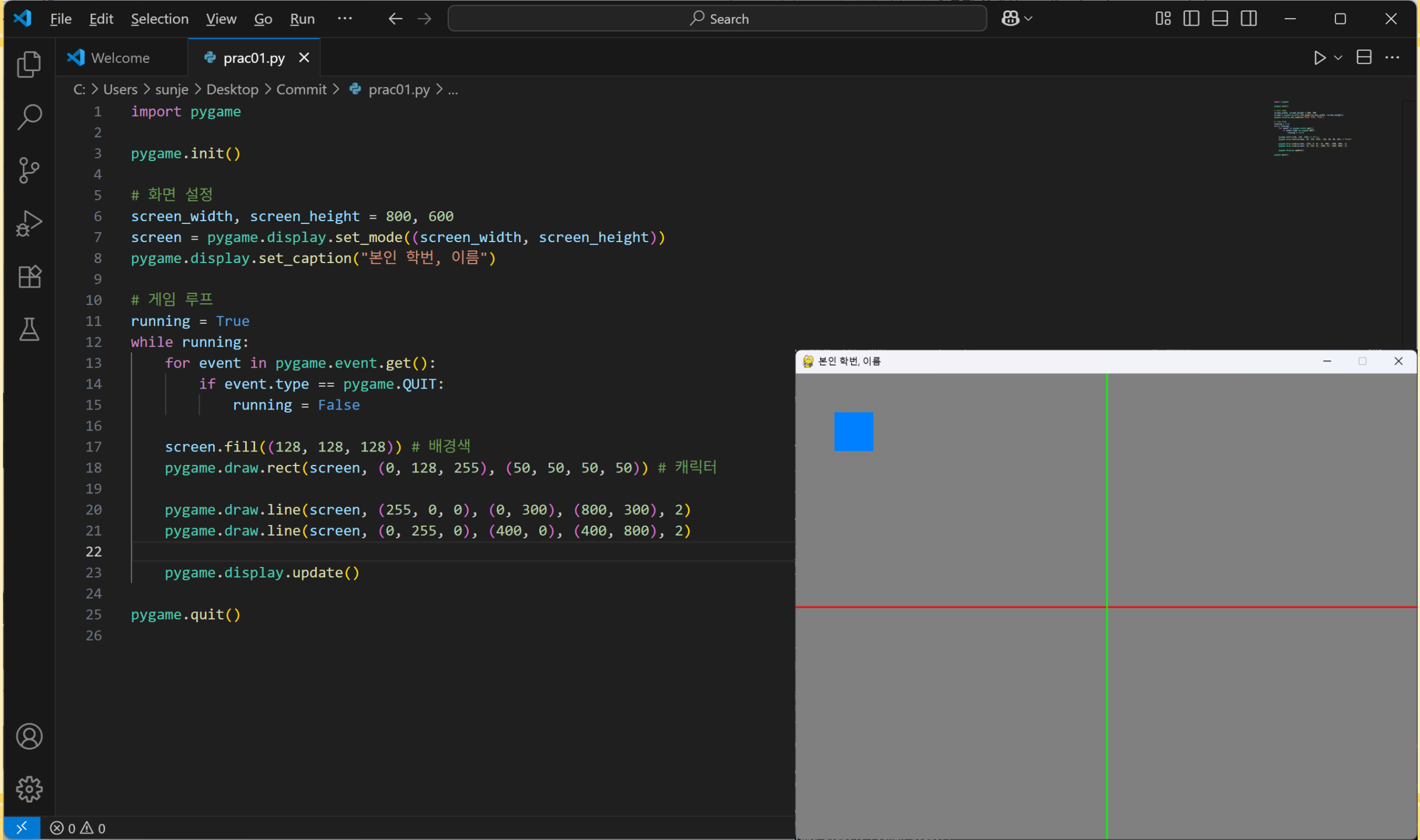
- 캐릭터를 다른 도형(+다른 색상)으로 변경하기
 - 예) 원, 삼각형, 오각형, 별

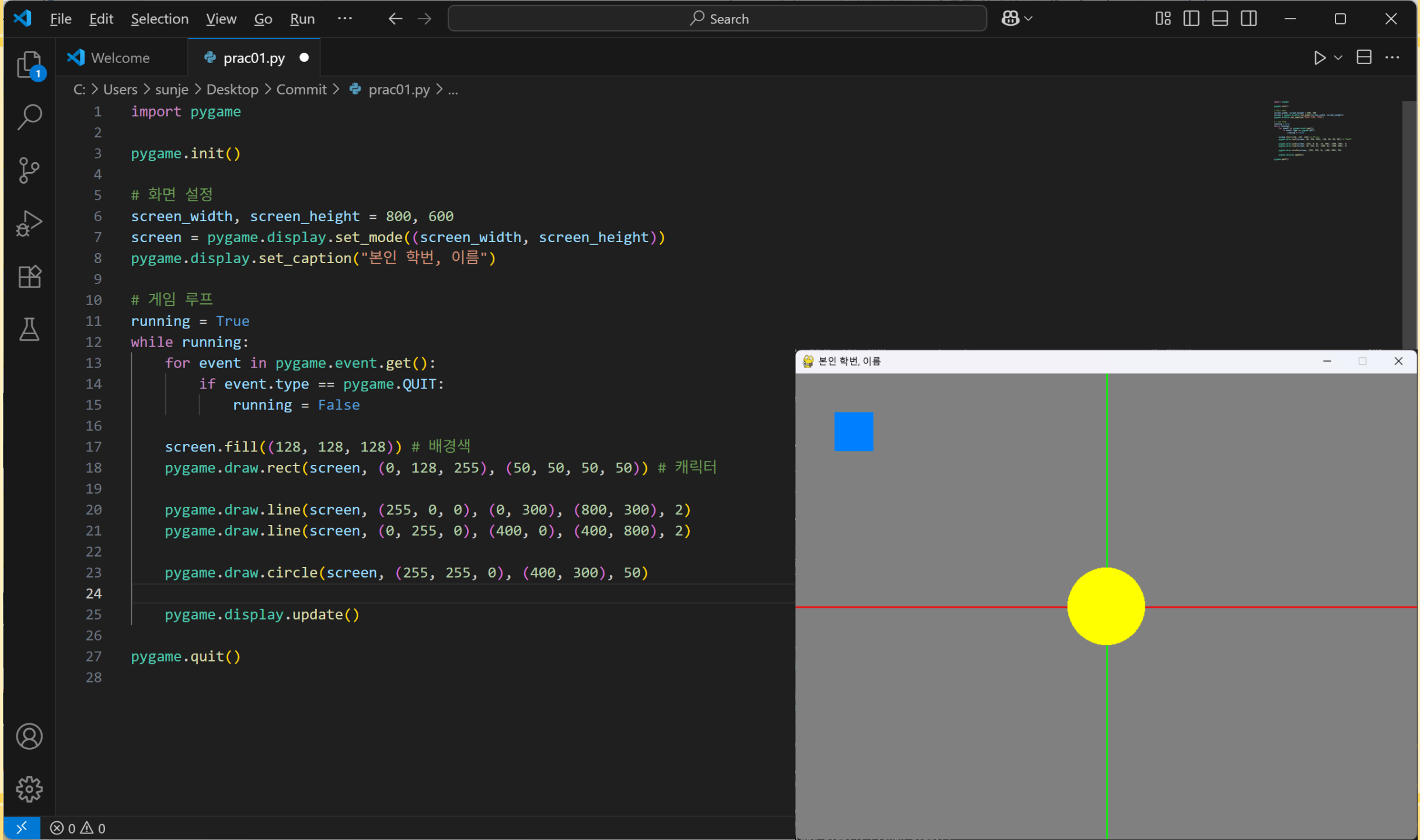
Welcome prac01.py X

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9
10 # 게임 루프
11 running = True
12 while running:
13     for event in pygame.event.get():
14         if event.type == pygame.QUIT:
15             running = False
16
17     screen.fill((128, 128, 128)) # 배경색
18     pygame.draw.rect(screen, (0, 128, 255), (50, 50, 50, 50), 5) # 캐릭터
19     pygame.display.update()
20
21 pygame.quit()
22
```

본인 학번, 이름







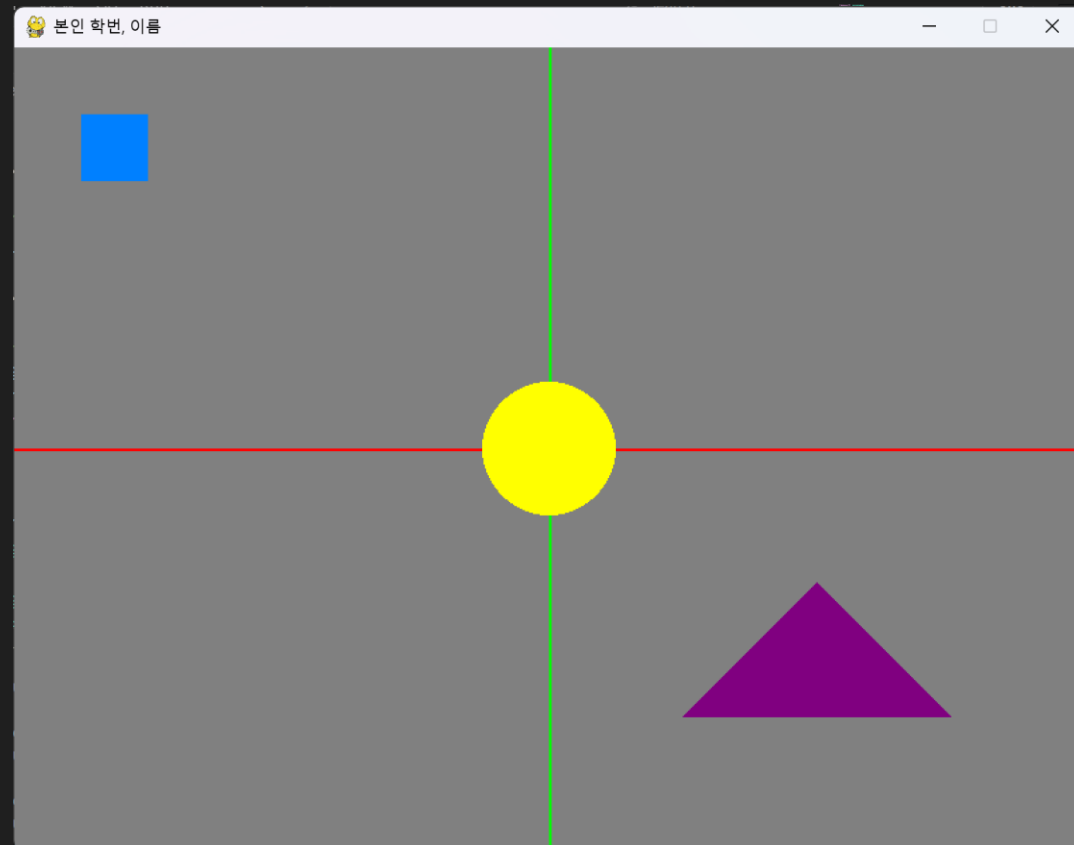
Welcome

prac01.py X

prac01-1.py 끝

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9
10 # 게임 루프
11 running = True
12 while running:
13     for event in pygame.event.get():
14         if event.type == pygame.QUIT:
15             running = False
16
17     screen.fill((128, 128, 128)) # 배경색
18     pygame.draw.rect(screen, (0, 128, 255), (50, 50, 50, 50)) # 캐릭터
19
20     pygame.draw.line(screen, (255, 0, 0), (0, 300), (800, 300), 2)
21     pygame.draw.line(screen, (0, 255, 0), (400, 0), (400, 800), 2)
22
23     pygame.draw.circle(screen, (255, 255, 0), (400, 300), 50)
24
25     pygame.draw.polygon(screen, (128, 0, 128), ((600, 400), (500, 500), (700, 500)))
26
27     pygame.display.update()
28
29 pygame.quit()
30
```



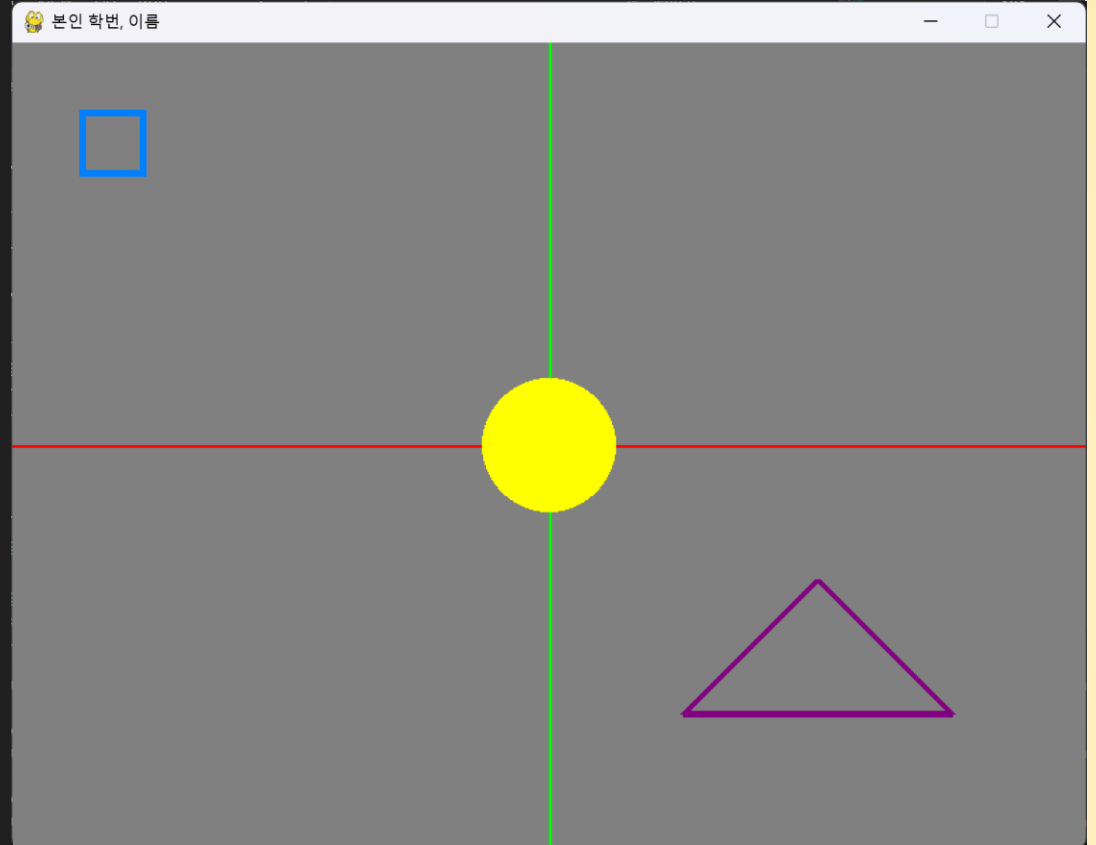


Welcome

prac01.py

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9
10 # 게임 루프
11 running = True
12 while running:
13     for event in pygame.event.get():
14         if event.type == pygame.QUIT:
15             running = False
16
17     screen.fill((128, 128, 128)) # 배경색
18     pygame.draw.rect(screen, (0, 128, 255), (50, 50, 50, 50), 5) # 캐릭터
19
20     pygame.draw.line(screen, (255, 0, 0), (255, 0, 0), (0, 300), (800, 300), 2)
21     pygame.draw.line(screen, (0, 255, 0), (400, 0), (400, 800), 2)
22
23     pygame.draw.circle(screen, (255, 255, 0), (400, 300), 50)
24
25     pygame.draw.polygon(screen, (128, 0, 128), ((600, 400), (500, 500), (700, 500)), 5)
26
27     pygame.display.update()
28
29 pygame.quit()
30
```

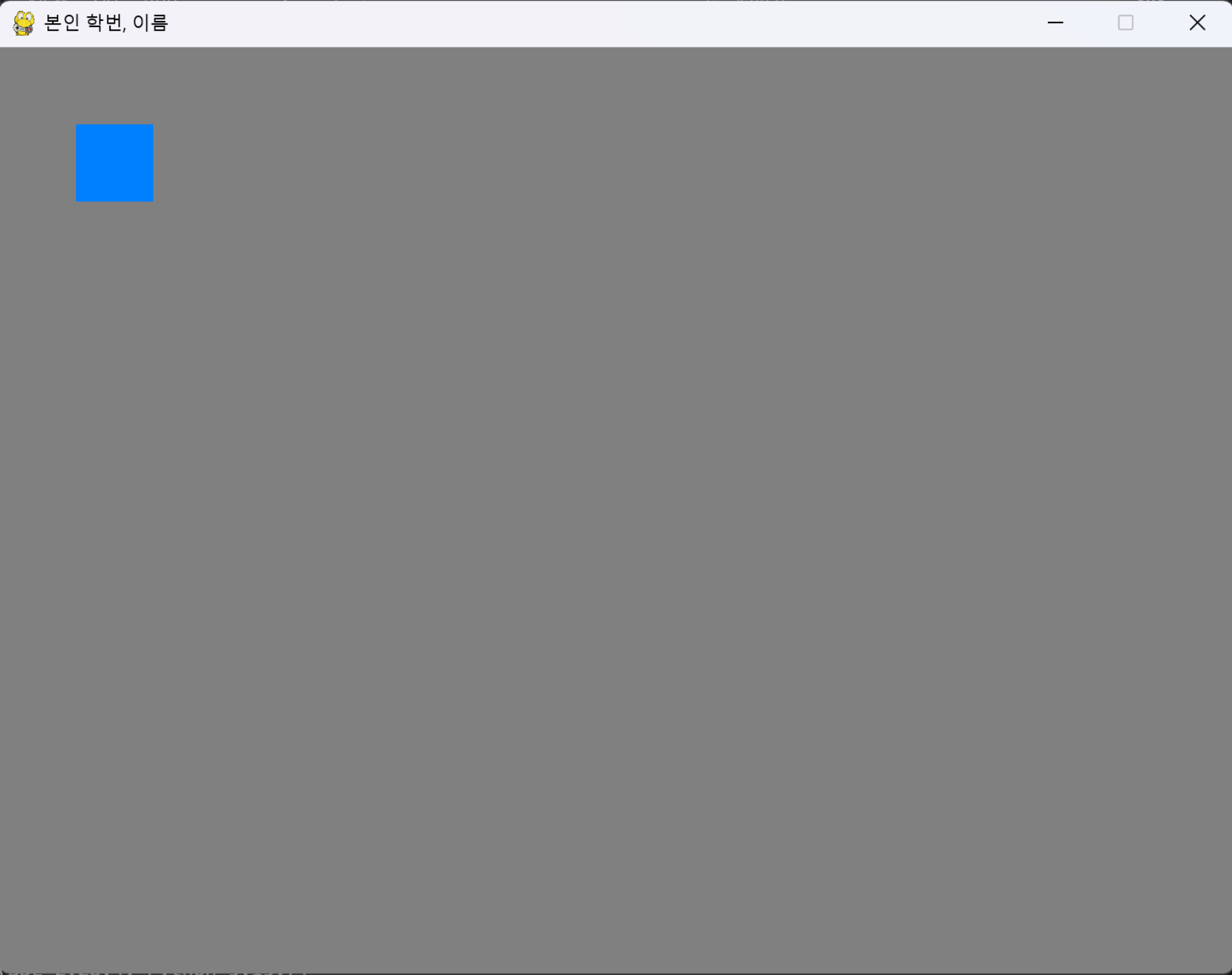


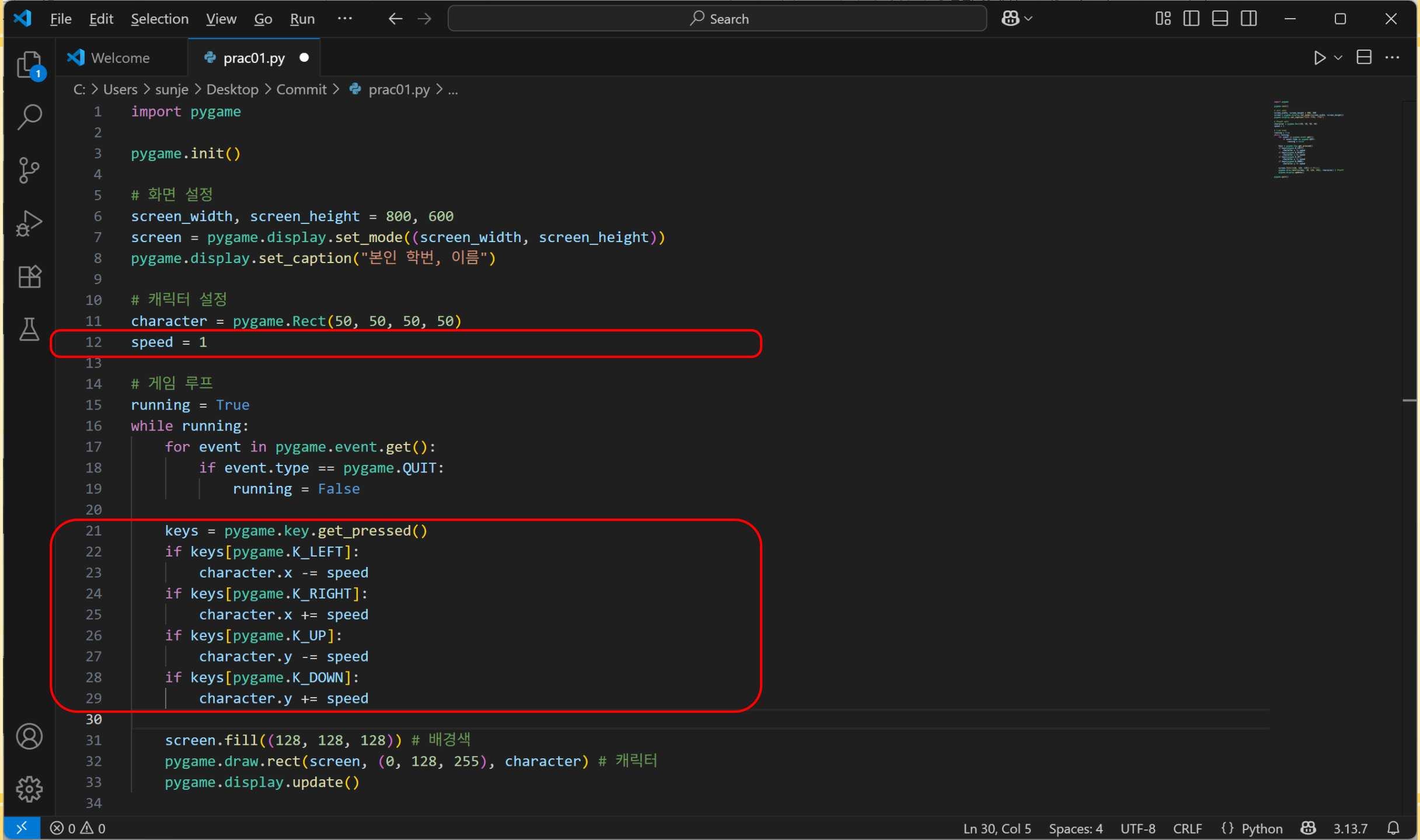
C: > Users > sunje > Desktop > Commit >  prac01.py > ...

```

1  import pygame
2
3  pygame.init()
4
5  # 화면 설정
6  screen_width, screen_height = 800, 600
7  screen = pygame.display.set_mode((screen_width, screen_height))
8  pygame.display.set_caption("본인 학번, 이름")
9
10 # 캐릭터 설정
11 character = pygame.Rect(50, 50, 50, 50)
12
13 # 게임 루프
14 running = True
15 while running:
16     for event in pygame.event.get():
17         if event.type == pygame.QUIT:
18             running = False
19
20     screen.fill((128, 128, 128)) # 배경색
21     pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
22     pygame.display.update()
23
24 pygame.quit()
25

```









Welcome

prac01.py

prac01-2.py

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
5  # 화면 설정
6  screen_width, screen_height = 800, 600
7  screen = pygame.display.set_mode((screen_width, screen_height))
8  pygame.display.set_caption("본인 학번, 이름")
9  clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 50, 50)
13 speed = 1
14
15 # 게임 루프
16 running = True
17 while running:
18     for event in pygame.event.get():
19         if event.type == pygame.QUIT:
20             running = False
21
22     keys = pygame.key.get_pressed()
23     if keys[pygame.K_LEFT]:
24         character.x -= speed
25     if keys[pygame.K_RIGHT]:
26         character.x += speed
27     if keys[pygame.K_UP]:
28         character.y -= speed
29     if keys[pygame.K_DOWN]:
30         character.y += speed
31
32     screen.fill((128, 128, 128)) # 배경색
33     pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
34     pygame.display.update()
35
36     clock.tick(60) # 60 FPS
37
38 pygame.quit()
```

```
pygame.init()
screen_width, screen_height = 800, 600
screen = pygame.display.set_mode((screen_width, screen_height))
pygame.display.set_caption("본인 학번, 이름")
clock = pygame.time.Clock()

# 캐릭터 설정
character = pygame.Rect(50, 50, 50, 50)
speed = 1

# 게임 루프
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        character.x -= speed
    if keys[pygame.K_RIGHT]:
        character.x += speed
    if keys[pygame.K_UP]:
        character.y -= speed
    if keys[pygame.K_DOWN]:
        character.y += speed

    screen.fill((128, 128, 128)) # 배경색
    pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
    pygame.display.update()

    clock.tick(60) # 60 FPS

pygame.quit()
```



연습 문제 (3)

- 도형 대신 이미지를 출력하기



<https://github.com/ProfSunKim/commit/>





Welcome | prac01.py

prac01-3.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```

11 # 캐릭터 설정
12 #character = pygame.Rect(50, 50, 50, 50)
13 speed = 1
14 charcter_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 posX, posY = 50, 50
16
17 # 게임 루프
18 running = True
19 while running:
20     for event in pygame.event.get():
21         if event.type == pygame.QUIT:
22             running = False
23
24     keys = pygame.key.get_pressed()
25     if keys[pygame.K_LEFT]:
26         #character.x -= speed
27         posX -= speed
28     if keys[pygame.K_RIGHT]:
29         #character.x += speed
30         posX += speed
31     if keys[pygame.K_UP]:
32         #character.y -= speed
33         posY -= speed
34     if keys[pygame.K_DOWN]:
35         #character.y += speed
36         posY += speed
37
38     screen.fill((128, 128, 128)) # 배경색
39     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
40     screen.blit(charcter_image, (posX, posY))
41     pygame.display.update()
42
43     clock.tick(60) # 60 FPS
44

```

컴퓨터마다
폴더 위치가 다름







Welcome

prac01.py

prac01-3.py 끝

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
11 # 캐릭터 설정
12 #character = pygame.Rect(50, 50, 50, 50)
13 speed = 1
14 charcter_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 charcter_position = [50, 50]
16
17 # 게임 루프
18 running = True
19 while running:
20     for event in pygame.event.get():
21         if event.type == pygame.QUIT:
22             running = False
23
24     keys = pygame.key.get_pressed()
25     if keys[pygame.K_LEFT]:
26         #character.x -= speed
27         charcter_position[0] -= speed
28     if keys[pygame.K_RIGHT]:
29         #character.x += speed
30         charcter_position[0] += speed
31     if keys[pygame.K_UP]:
32         #character.y -= speed
33         charcter_position[1] -= speed
34     if keys[pygame.K_DOWN]:
35         #character.y += speed
36         charcter_position[1] += speed
37
38     screen.fill((128, 128, 128)) # 배경색
39     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
40     screen.blit(charcter_image, charcter_position)
41     pygame.display.update()
42
43     clock.tick(60) # 60 FPS
44
```

```
1 import pygame
2 import sys
3
4 # 초기화
5 pygame.init()
6 screen = pygame.display.set_mode((128, 128))
7 pygame.display.set_caption('pygame test')
8
9 # 캐릭터 설정
10 #character = pygame.Rect(50, 50, 50, 50)
11 speed = 1
12 charcter_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
13 charcter_position = [50, 50]
14
15 # 게임 루프
16 running = True
17 while running:
18     for event in pygame.event.get():
19         if event.type == pygame.QUIT:
20             running = False
21
22     keys = pygame.key.get_pressed()
23     if keys[pygame.K_LEFT]:
24         #character.x -= speed
25         charcter_position[0] -= speed
26     if keys[pygame.K_RIGHT]:
27         #character.x += speed
28         charcter_position[0] += speed
29     if keys[pygame.K_UP]:
30         #character.y -= speed
31         charcter_position[1] -= speed
32     if keys[pygame.K_DOWN]:
33         #character.y += speed
34         charcter_position[1] += speed
35
36     screen.fill((128, 128, 128)) # 배경색
37     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
38     screen.blit(charcter_image, charcter_position)
39     pygame.display.update()
40
41     clock.tick(60) # 60 FPS
42
```

연습 문제 (4)

- 각자 원하는 그림으로 변경하기

게임 제작 (2)

1. 화면 구성 및 캐릭터 이동
2. 맵 생성 및 충돌 처리
 - 타일 기반 맵 생성 – 타일을 사용하여 맵을 생성 (맵 데이터 파일 로드)
 - 충돌 처리 – 캐릭터가 맵의 벽과 충돌하지 않도록 충돌 감지 및 반응을 구현
3. 적 생성 및 전투 시스템
4. 아이템 생성 및 게임 완성



Welcome

prac01.py X

prac01-4.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 맵 설정
18 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
19
20 # 게임 루프
21 running = True
22 while running:
23     for event in pygame.event.get():
24         if event.type == pygame.QUIT:
25             running = False
26
27     keys = pygame.key.get_pressed()
28     if keys[pygame.K_LEFT]:
29         #character.x -= speed
30         character_position[0] -= speed
31     if keys[pygame.K_RIGHT]:
32         #character.x += speed
33         character_position[0] += speed
34     if keys[pygame.K_UP]:
35         #character.y -= speed
36         character_position[1] -= speed
37     if keys[pygame.K_DOWN]:
38         #character.y += speed
39         character_position[1] += speed
40
41     screen.fill((128, 128, 128)) # 배경색
42
43     for wall in walls:
44         pygame.draw.rect(screen, (128, 0, 0), wall)
45
46     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
```

```
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 맵 설정
18 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
19
20 # 게임 루프
21 running = True
22 while running:
23     for event in pygame.event.get():
24         if event.type == pygame.QUIT:
25             running = False
26
27     keys = pygame.key.get_pressed()
28     if keys[pygame.K_LEFT]:
29         #character.x -= speed
30         character_position[0] -= speed
31     if keys[pygame.K_RIGHT]:
32         #character.x += speed
33         character_position[0] += speed
34     if keys[pygame.K_UP]:
35         #character.y -= speed
36         character_position[1] -= speed
37     if keys[pygame.K_DOWN]:
38         #character.y += speed
39         character_position[1] += speed
40
41     screen.fill((128, 128, 128)) # 배경색
42
43     for wall in walls:
44         pygame.draw.rect(screen, (128, 0, 0), wall)
45
46     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
```



연습 문제 (5)

- 벽(타일) 하나 더 추가하기

 본인 학번, 이름





Welcome

prac01.py



C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1  import pygame
2
3  pygame.init()
4
5  # 화면 설정
6  screen_width, screen_height = 800, 600
7  screen = pygame.display.set_mode((screen_width, screen_height))
8  pygame.display.set_caption("본인 학번, 이름")
9  clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 #character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 #character_position = [50, 50]
16
17 # 맵 설정
18 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
19
20 # 게임 루프
21 running = True
22 while running:
23     for event in pygame.event.get():
24         if event.type == pygame.QUIT:
25             running = False
26
27     keys = pygame.key.get_pressed()
28     if keys[pygame.K_LEFT]:
29         character.x -= speed
30         #character_position[0] -= speed
31     if keys[pygame.K_RIGHT]:
32         character.x += speed
33         #character_position[0] += speed
34     if keys[pygame.K_UP]:
```

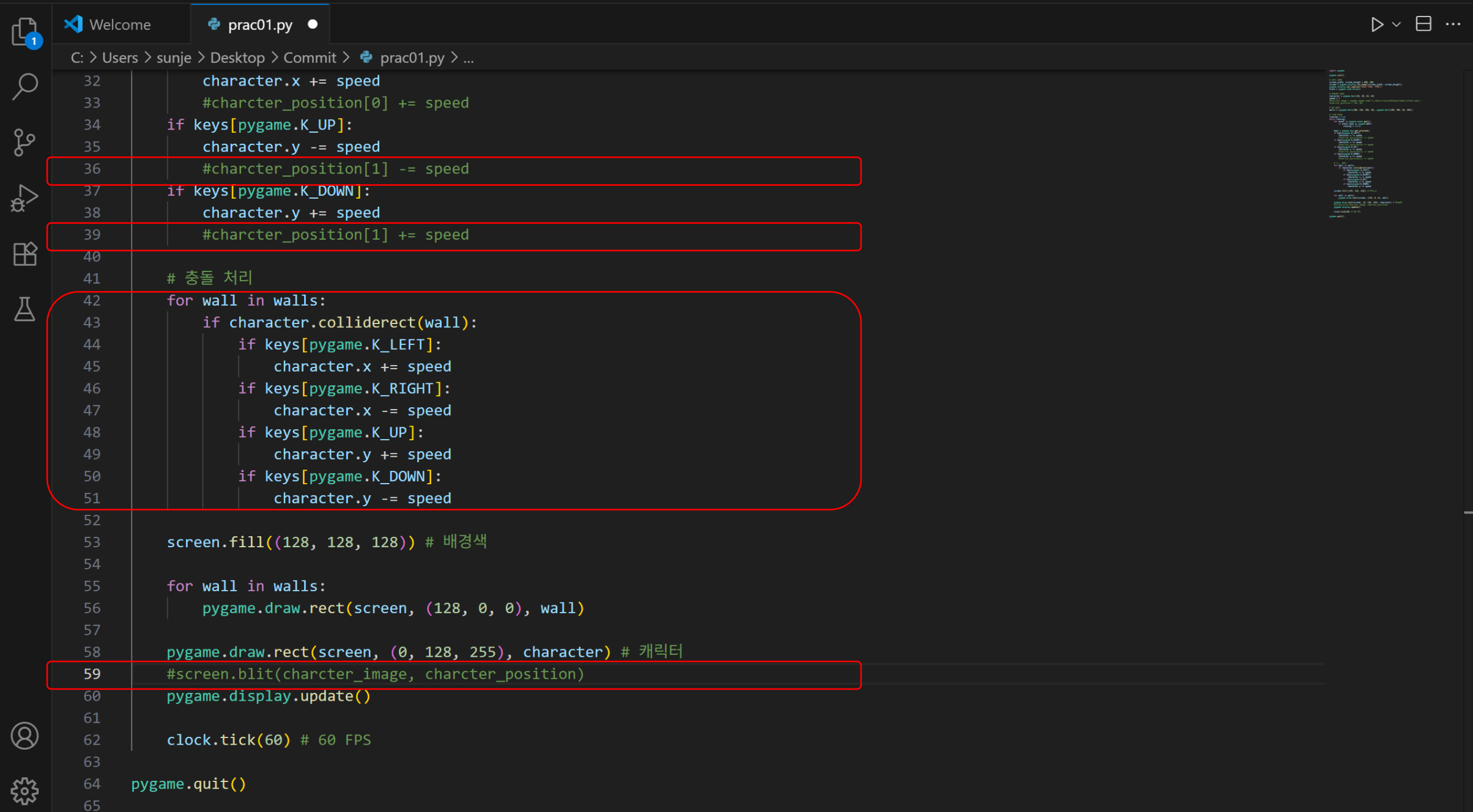
```
pygame.init()
screen_width, screen_height = 800, 600
screen = pygame.display.set_mode((screen_width, screen_height))
pygame.display.set_caption("본인 학번, 이름")
clock = pygame.time.Clock()

# 캐릭터 설정
character = pygame.Rect(50, 50, 64, 64)
speed = 1
#character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
#character_position = [50, 50]

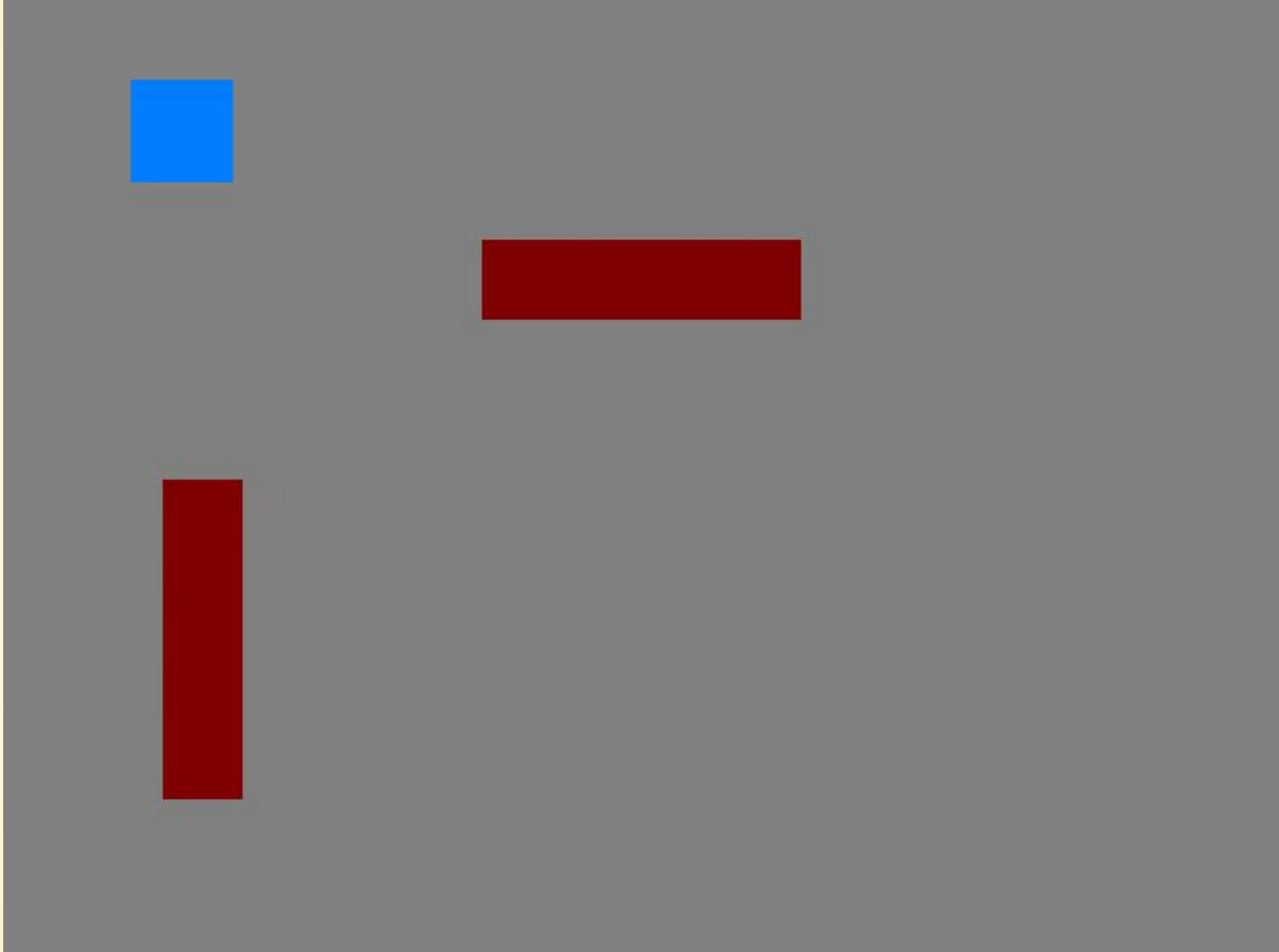
# 맵 설정
walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]

# 게임 루프
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        character.x -= speed
        #character_position[0] -= speed
    if keys[pygame.K_RIGHT]:
        character.x += speed
        #character_position[0] += speed
    if keys[pygame.K_UP]:
```



```
32 character.x += speed
33 #character_position[0] += speed
34 if keys[pygame.K_UP]:
35     character.y -= speed
36     #character_position[1] -= speed
37 if keys[pygame.K_DOWN]:
38     character.y += speed
39     #character_position[1] += speed
40
41 # 충돌 처리
42 for wall in walls:
43     if character.colliderect(wall):
44         if keys[pygame.K_LEFT]:
45             character.x += speed
46         if keys[pygame.K_RIGHT]:
47             character.x -= speed
48         if keys[pygame.K_UP]:
49             character.y += speed
50         if keys[pygame.K_DOWN]:
51             character.y -= speed
52
53 screen.fill((128, 128, 128)) # 배경색
54
55 for wall in walls:
56     pygame.draw.rect(screen, (128, 0, 0), wall)
57
58 pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
59 #screen.blit(charcter_image, charcter_position)
60 pygame.display.update()
61
62 clock.tick(60) # 60 FPS
63
64 pygame.quit()
65
```



연습 문제 (6)

- 각자 원하는 그림으로 변경하기
- 주의!!! 충돌 처리는 Rect에서만 가능하기 때문에, 이미지를 Rect 위에 그려야 함



Welcome

prac01.py

C: > Users > sunje > Desktop > Commit > prac01.py > [charcter_image]

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 맵 설정
18 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
19
20 # 게임 루프
21 running = True
22 while running:
23     for event in pygame.event.get():
24         if event.type == pygame.QUIT:
25             running = False
26
27     keys = pygame.key.get_pressed()
28     if keys[pygame.K_LEFT]:
29         character.x -= speed
30         character_position[0] -= speed
31     if keys[pygame.K_RIGHT]:
32         character.x += speed
33         character_position[0] += speed
34     if keys[pygame.K_UP]:
```

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 맵 설정
18 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
19
20 # 게임 루프
21 running = True
22 while running:
23     for event in pygame.event.get():
24         if event.type == pygame.QUIT:
25             running = False
26
27     keys = pygame.key.get_pressed()
28     if keys[pygame.K_LEFT]:
29         character.x -= speed
30         character_position[0] -= speed
31     if keys[pygame.K_RIGHT]:
32         character.x += speed
33         character_position[0] += speed
34     if keys[pygame.K_UP]:
```



Welcome

prac01.py

prac01-4.py



C: > Users > sunje > Desktop > Commit > prac01.py > [charcter_image]

```
34     if keys[pygame.K_UP]:
35         character.y -= speed
36         charcter_position[1] -= speed
37     if keys[pygame.K_DOWN]:
38         character.y += speed
39         charcter_position[1] += speed
40
41     # 충돌 처리
42     for wall in walls:
43         if character.colliderect(wall):
44             if keys[pygame.K_LEFT]:
45                 character.x += speed
46                 charcter_position[0] += speed
47             if keys[pygame.K_RIGHT]:
48                 character.x -= speed
49                 charcter_position[0] -= speed
50             if keys[pygame.K_UP]:
51                 character.y += speed
52                 charcter_position[1] += speed
53             if keys[pygame.K_DOWN]:
54                 character.y -= speed
55                 charcter_position[1] -= speed
56
57     screen.fill((128, 128, 128)) # 배경색
58
59     for wall in walls:
60         pygame.draw.rect(screen, (128, 0, 0), wall)
61
62     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
63     screen.blit(charcter_image, charcter_position)
64     pygame.display.update()
65
66     clock.tick(60) # 60 FPS
67
```

```
pygame.init()
screen = pygame.display.set_mode((1280, 720))
pygame.display.set_caption("prac01-4.py")

# 캐릭터 이미지 로드
charcter_image = pygame.image.load('character.png')

# 벽 이미지 로드
wall_image = pygame.image.load('wall.png')

# 캐릭터 초기 위치
character_x = 50
character_y = 50
character_x2 = 150
character_y2 = 150

# 벽 초기 위치
wall_x = 100
wall_y = 100
wall_x2 = 200
wall_y2 = 200

# 속도
speed = 5

# 방향키
keys = pygame.key.get_pressed()

# 게임 루프
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            sys.exit()

    # 키보드 입력 처리
    if keys[pygame.K_UP]:
        character_y -= speed
    elif keys[pygame.K_DOWN]:
        character_y += speed
    elif keys[pygame.K_LEFT]:
        character_x -= speed
    elif keys[pygame.K_RIGHT]:
        character_x += speed

    # 충돌 처리
    if character.colliderect(wall):
        if keys[pygame.K_LEFT]:
            character_x += speed
        elif keys[pygame.K_RIGHT]:
            character_x -= speed
        elif keys[pygame.K_UP]:
            character_y += speed
        elif keys[pygame.K_DOWN]:
            character_y -= speed

    # 화면 업데이트
    screen.fill((128, 128, 128))
    pygame.draw.rect(screen, (128, 0, 0), wall)
    #pygame.draw.rect(screen, (0, 128, 255), character)
    screen.blit(charcter_image, (character_x, character_y))
    pygame.display.update()

    clock.tick(60)
```



게임 제작 (3)

1. 화면 구성 및 캐릭터 이동
2. 맵 생성 및 충돌 처리
3. 적 생성 및 전투 시스템
 - 적 캐릭터 생성 – 간단한 AI를 가진 적 캐릭터를 생성하고 맵에 배치
 - 전투 시스템 구현 – 플레이어와 적 간의 기본적인 전투 메커니즘을 구현 (예: 공격력, 체력 등)
4. 아이템 생성 및 게임 완성



Welcome

prac01.py

prac01-5.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20
21 # 맵 설정
22 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
23
24 # 게임 루프
25 running = True
26 while running:
27     for event in pygame.event.get():
28         if event.type == pygame.QUIT:
29             running = False
30
31     keys = pygame.key.get_pressed()
32     if keys[pygame.K_LEFT]:
33         character.x -= speed
34         character_position[0] -= speed
```

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20
21 # 맵 설정
22 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
23
24 # 게임 루프
25 running = True
26 while running:
27     for event in pygame.event.get():
28         if event.type == pygame.QUIT:
29             running = False
30
31     keys = pygame.key.get_pressed()
32     if keys[pygame.K_LEFT]:
33         character.x -= speed
34         character_position[0] -= speed
```



Welcome

prac01.py

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
41     if keys[pygame.K_DOWN]:
42         character.y += speed
43         charcter_position[1] += speed
44
45     # 충돌 처리
46     for wall in walls:
47         if character.colliderect(wall):
48             if keys[pygame.K_LEFT]:
49                 character.x += speed
50                 charcter_position[0] += speed
51             if keys[pygame.K_RIGHT]:
52                 character.x -= speed
53                 charcter_position[0] -= speed
54             if keys[pygame.K_UP]:
55                 character.y += speed
56                 charcter_position[1] += speed
57             if keys[pygame.K_DOWN]:
58                 character.y -= speed
59                 charcter_position[1] -= speed
60
61     screen.fill((128, 128, 128)) # 배경색
62
63     for wall in walls:
64         pygame.draw.rect(screen, (128, 0, 0), wall)
65
66     pygame.draw.rect(screen, enemy_color, enemy)
67     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
68     screen.blit(charcter_image, charcter_position)
69     pygame.display.update()
70
71     clock.tick(60) # 60 FPS
72
73     pygame.quit()
74
```

```
import pygame
import sys
import random

# 초기화
pygame.init()
screen = pygame.display.set_mode((1280, 720))
pygame.display.set_caption("pygame practice")

# 캐릭터 설정
character = pygame.Rect(50, 50, 50, 50)
character_color = (0, 128, 255)
speed = 5

# 벽 설정
walls = []
for i in range(10):
    wall = pygame.Rect(random.randint(0, 1280), random.randint(0, 720), 100, 100)
    walls.append(wall)

# 배경색
background_color = (128, 128, 128)

# 캐릭터 이미지
charcter_image = pygame.image.load('character.png')

# enemy 설정
enemy_color = (128, 0, 0)
enemy = pygame.Rect(100, 100, 50, 50)

# 게임 루프
clock = pygame.time.Clock()
while True:
    # 이벤트 처리
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            sys.exit()

    # 키보드 입력
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        character.x -= speed
    elif keys[pygame.K_RIGHT]:
        character.x += speed
    elif keys[pygame.K_UP]:
        character.y -= speed
    elif keys[pygame.K_DOWN]:
        character.y += speed

    # 충돌 처리
    for wall in walls:
        if character.colliderect(wall):
            if keys[pygame.K_LEFT]:
                character.x += speed
            elif keys[pygame.K_RIGHT]:
                character.x -= speed
            elif keys[pygame.K_UP]:
                character.y += speed
            elif keys[pygame.K_DOWN]:
                character.y -= speed

    # 화면 업데이트
    screen.fill(background_color)

    for wall in walls:
        pygame.draw.rect(screen, (128, 0, 0), wall)

    pygame.draw.rect(screen, enemy_color, enemy)

    #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
    screen.blit(charcter_image, character.position)

    pygame.display.update()

    clock.tick(60) # 60 FPS

pygame.quit()
```





Welcome

prac01.py

prac01-5.py 끝



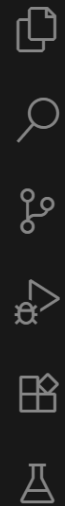
C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
45     # 충돌 처리
46     for wall in walls:
47         if character.colliderect(wall):
48             if keys[pygame.K_LEFT]:
49                 character.x += speed
50                 charcter_position[0] += speed
51             if keys[pygame.K_RIGHT]:
52                 character.x -= speed
53                 charcter_position[0] -= speed
54             if keys[pygame.K_UP]:
55                 character.y += speed
56                 charcter_position[1] += speed
57             if keys[pygame.K_DOWN]:
58                 character.y -= speed
59                 charcter_position[1] -= speed
60
61     # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
62     if character.colliderect(enemy):
63         enemy_color = (255, 255, 0)
64
65     screen.fill((128, 128, 128)) # 배경색
66
67     for wall in walls:
68         pygame.draw.rect(screen, (128, 0, 0), wall)
69
70     pygame.draw.rect(screen, enemy_color, enemy)
71     #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
72     screen.blit(charcter_image, charcter_position)
73     pygame.display.update()
74
75     clock.tick(60) # 60 FPS
76
77 pygame.quit()
78
```

```
1 import pygame
2 import sys
3
4 # 초기화
5 pygame.init()
6 screen = pygame.display.set_mode((800, 600))
7 pygame.display.set_caption("prac01-5.py")
8
9 # 캐릭터 설정
10 character_x = 50
11 character_y = 50
12 character_width = 50
13 character_height = 50
14
15 # 벽 설정
16 walls = []
17 walls.append(pygame.Rect(100, 100, 100, 100))
18 walls.append(pygame.Rect(200, 100, 100, 100))
19 walls.append(pygame.Rect(300, 100, 100, 100))
20 walls.append(pygame.Rect(400, 100, 100, 100))
21 walls.append(pygame.Rect(500, 100, 100, 100))
22 walls.append(pygame.Rect(600, 100, 100, 100))
23 walls.append(pygame.Rect(700, 100, 100, 100))
24 walls.append(pygame.Rect(800, 100, 100, 100))
25
26 # 적 설정
27 enemy_x = 750
28 enemy_y = 50
29 enemy_width = 50
30 enemy_height = 50
31
32 # 배경 설정
33 background_color = (128, 128, 128)
34
35 # 캐릭터 이미지
36 charcter_image = pygame.image.load('character.png')
37
38 # 게임 루프
39 while True:
40     # 이벤트 처리
41     for event in pygame.event.get():
42         if event.type == pygame.QUIT:
43             pygame.quit()
44             sys.exit()
45
46     # 키 처리
47     keys = pygame.key.get_pressed()
48
49     # 충돌 처리
50     for wall in walls:
51         if character.colliderect(wall):
52             if keys[pygame.K_LEFT]:
53                 character_x += speed
54                 charcter_position[0] += speed
55             if keys[pygame.K_RIGHT]:
56                 character_x -= speed
57                 charcter_position[0] -= speed
58             if keys[pygame.K_UP]:
59                 character_y += speed
60                 charcter_position[1] += speed
61             if keys[pygame.K_DOWN]:
62                 character_y -= speed
63                 charcter_position[1] -= speed
64
65     # 전투 시스템
66     if character.colliderect(enemy):
67         enemy_color = (255, 255, 0)
68
69     # 화면 업데이트
70     screen.fill(background_color)
71
72     for wall in walls:
73         pygame.draw.rect(screen, (128, 0, 0), wall)
74
75     pygame.draw.rect(screen, enemy_color, enemy)
76     #pygame.draw.rect(screen, (0, 128, 255), character)
77     screen.blit(charcter_image, charcter_position)
78     pygame.display.update()
79
80     # FPS 설정
81     clock.tick(60)
82
83 pygame.quit()
```



※ 화면 캡처를 위해 enemy 위치를 변경함



Welcome

prac01.py X

prac01-6.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > [en] enemy_speed

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = -1
21
22 # 맵 설정
23 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
24
25 # 게임 루프
26 running = True
27 while running:
28     for event in pygame.event.get():
29         if event.type == pygame.QUIT:
30             running = False
31
32     keys = pygame.key.get_pressed()
33     if keys[pygame.K_LEFT]:
34         character.x -= speed
```

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = -1
21
22 # 맵 설정
23 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
24
25 # 게임 루프
26 running = True
27 while running:
28     for event in pygame.event.get():
29         if event.type == pygame.QUIT:
30             running = False
31
32     keys = pygame.key.get_pressed()
33     if keys[pygame.K_LEFT]:
34         character.x -= speed
```





Welcome

prac01.py X

C: > Users > sunje > Desktop > Commit > prac01.py > enemy_speed

```
46 # 충돌 처리
47 for wall in walls:
48     if character.colliderect(wall):
49         if keys[pygame.K_LEFT]:
50             character.x += speed
51             charcter_position[0] += speed
52         if keys[pygame.K_RIGHT]:
53             character.x -= speed
54             charcter_position[0] -= speed
55         if keys[pygame.K_UP]:
56             character.y += speed
57             charcter_position[1] += speed
58         if keys[pygame.K_DOWN]:
59             character.y -= speed
60             charcter_position[1] -= speed
61
62 # 적 이동
63 enemy.x += enemy_speed
64 if enemy.x < 400:
65     enemy_speed = 1
66 if enemy.x > 600:
67     enemy_speed = -1
68
69 # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
70 if character.colliderect(enemy):
71     enemy_color = (255, 255, 0)
72
73 screen.fill((128, 128, 128)) # 배경색
74
75 for wall in walls:
76     pygame.draw.rect(screen, (128, 0, 0), wall)
77
78 pygame.draw.rect(screen, enemy_color, enemy)
79 #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
```

```
import pygame
import sys
import random

# 초기화
pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Enemy Speed Control")

# 캐릭터 설정
character_x = 50
character_y = 50
character_width = 50
character_height = 50
character_color = (0, 128, 255)

# 벽 설정
walls = []
for i in range(10):
    wall_x = random.randint(0, 750)
    wall_y = random.randint(0, 550)
    wall_width = random.randint(50, 100)
    wall_height = random.randint(50, 100)
    walls.append(pygame.Rect(wall_x, wall_y, wall_width, wall_height))

# 적 설정
enemy_x = 500
enemy_y = 500
enemy_width = 50
enemy_height = 50
enemy_color = (255, 255, 0)

# 속도 설정
speed = 5
enemy_speed = 1

# 게임 루프
running = True
while running:
    # 이벤트 처리
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # 키 처리
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        character_x -= speed
    if keys[pygame.K_RIGHT]:
        character_x += speed
    if keys[pygame.K_UP]:
        character_y -= speed
    if keys[pygame.K_DOWN]:
        character_y += speed

    # 적 이동
    enemy_x += enemy_speed
    if enemy_x < 400:
        enemy_speed = 1
    if enemy_x > 600:
        enemy_speed = -1

    # 충돌 감지
    if character.colliderect(enemy):
        enemy_color = (255, 255, 0)

    # 화면 업데이트
    screen.fill((128, 128, 128))
    for wall in walls:
        pygame.draw.rect(screen, (128, 0, 0), wall)
    pygame.draw.rect(screen, enemy_color, enemy)
    #pygame.draw.rect(screen, (0, 128, 255), character)

    pygame.display.flip()

pygame.quit()
sys.exit()
```




연습 문제 (7)

- 적(enemy)의 움직임이 위/아래(y축)로 함께 움직이도록 만들어 보기





Welcome

prac01.py X

C: > Users > sunje > Desktop > Commit > prac01.py > [en] enemy_speed

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = [-1, -1]
21
22 # 맵 설정
23 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
24
25 # 게임 루프
26 running = True
27 while running:
28     for event in pygame.event.get():
29         if event.type == pygame.QUIT:
30             running = False
31
32     keys = pygame.key.get_pressed()
33     if keys[pygame.K_LEFT]:
34         character.x -= speed
```

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = [-1, -1]
21
22 # 맵 설정
23 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
24
25 # 게임 루프
26 running = True
27 while running:
28     for event in pygame.event.get():
29         if event.type == pygame.QUIT:
30             running = False
31
32     keys = pygame.key.get_pressed()
33     if keys[pygame.K_LEFT]:
34         character.x -= speed
```



Welcome

prac01.py X

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
45
46     # 충돌 처리
47     for wall in walls:
48         if character.collidect(wall):
49             if keys[pygame.K_LEFT]:
50                 character.x += speed
51                 charcter_position[0] += speed
52             if keys[pygame.K_RIGHT]:
53                 character.x -= speed
54                 charcter_position[0] -= speed
55             if keys[pygame.K_UP]:
56                 character.y += speed
57                 charcter_position[1] += speed
58             if keys[pygame.K_DOWN]:
59                 character.y -= speed
60                 charcter_position[1] -= speed
61
62     # 적 이동
63     enemy.x += enemy_speed[0]
64     if enemy.x < 400:
65         enemy_speed[0] = 1
66     if enemy.x > 600:
67         enemy_speed[0] = -1
68     enemy.y += enemy_speed[1]
69     if enemy.y < 300:
70         enemy_speed[1] = 1
71     if enemy.y > 500:
72         enemy_speed[1] = -1
73
74     # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
75     if character.collidect(enemy):
76         enemy_color = (255, 255, 0)
77
78     screen.fill((128, 128, 128)) # 배경색
```

```
import pygame
import sys
import random

# 초기화
pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Enemy Movement")

# 캐릭터 설정
character_x = 50
character_y = 50
character_speed = 5

# 벽 설정
walls = []
for i in range(10):
    wall_x = random.randint(100, 700)
    wall_y = random.randint(100, 500)
    walls.append([wall_x, wall_y])

# 적 설정
enemy_x = 500
enemy_y = 500
enemy_speed = [0, 0]

# 게임 루프
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            sys.exit()

    # 키 입력 처리
    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        character_x -= character_speed
    if keys[pygame.K_RIGHT]:
        character_x += character_speed
    if keys[pygame.K_UP]:
        character_y -= character_speed
    if keys[pygame.K_DOWN]:
        character_y += character_speed

    # 적 이동
    enemy_x += enemy_speed[0]
    enemy_y += enemy_speed[1]

    # 벽 충돌 처리
    for wall in walls:
        if character_x == wall[0] and character_y == wall[1]:
            # 충돌 처리
            pass

    # 적 이동 방향 설정
    if enemy_x < 400:
        enemy_speed[0] = 1
    if enemy_x > 600:
        enemy_speed[0] = -1
    if enemy_y < 300:
        enemy_speed[1] = 1
    if enemy_y > 500:
        enemy_speed[1] = -1

    # 화면 업데이트
    screen.fill((128, 128, 128))
    pygame.draw.rect(screen, (255, 255, 0), pygame.Rect(character_x, character_y, 50, 50))
    pygame.draw.rect(screen, (255, 255, 0), pygame.Rect(enemy_x, enemy_y, 50, 50))
    pygame.display.flip()

    # FPS 제한
    pygame.time.Clock().tick(60)
```



연습 문제 (8)

- 플레이어와 적이 충돌하면, 적의 움직임 멈추기



VS Code: Welcome | prac01.py X

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```

46 # 충돌 처리
47 for wall in walls:
48     if character.colliderect(wall):
49         if keys[pygame.K_LEFT]:
50             character.x += speed
51             charcter_position[0] += speed
52         if keys[pygame.K_RIGHT]:
53             character.x -= speed
54             charcter_position[0] -= speed
55         if keys[pygame.K_UP]:
56             character.y += speed
57             charcter_position[1] += speed
58         if keys[pygame.K_DOWN]:
59             character.y -= speed
60             charcter_position[1] -= speed
61
62 # 적 이동
63 enemy.x += enemy_speed[0]
64 if enemy.x < 400:
65     enemy_speed[0] = 1
66 if enemy.x > 600:
67     enemy_speed[0] = -1
68 enemy.y += enemy_speed[1]
69 if enemy.y < 300:
70     enemy_speed[1] = 1
71 if enemy.y > 500:
72     enemy_speed[1] = -1
73
74 # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
75 if character.colliderect(enemy):
76     enemy_color = (255, 255, 0)
77     enemy.x -= enemy_speed[0]
78     enemy.y -= enemy_speed[1]
79

```

[illegible]

게임 제작 (4)

1. 화면 구성 및 캐릭터 이동

2. 맵 생성 및 충돌 처리

3. 적 생성 및 전투 시스템

4. **아이템 생성 및 게임 완성**

- **아이템 생성 및 사용** – 게임 내에서 사용할 수 있는 아이템을 생성하고, 플레이어가 아이템을 획득하고 사용할 수 있도록 구현
- **게임 완성 및 테스트** – 게임을 테스트하고 버그를 수정하여 완성 (게임의 난이도 밸런싱으로 조정 가능)



Welcome

prac01.py

prac01-7.py 시작

C: > Users > sunje > Desktop > Commit > prac01.py > [0] items

```
1  import pygame
2
3  pygame.init()
4
5  # 화면 설정
6  screen_width, screen_height = 800, 600
7  screen = pygame.display.set_mode((screen_width, screen_height))
8  pygame.display.set_caption("본인 학번, 이름")
9  clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = [-1, -1]
21
22 # 아이템 설정
23 items = [pygame.Rect(200, 200, 20, 20), pygame.Rect(600, 200, 20, 20), pygame.Rect(200, 400, 20, 20)]
24
25 # 맵 설정
26 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
27
28 # 게임 루프
29 running = True
30 while running:
31     for event in pygame.event.get():
32         if event.type == pygame.QUIT:
33             running = False
34
```

```
'''
pygame
pygame.init()
pygame.display.set_mode((800, 600))
pygame.display.set_caption("본인 학번, 이름")
clock = pygame.time.Clock()

# 캐릭터 설정
character = pygame.Rect(50, 50, 64, 64)
speed = 1
character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
character_position = [50, 50]

# 적 설정
enemy = pygame.Rect(600, 400, 64, 64)
enemy_color = (255, 0, 0)
enemy_speed = [-1, -1]

# 아이템 설정
items = [pygame.Rect(200, 200, 20, 20), pygame.Rect(600, 200, 20, 20), pygame.Rect(200, 400, 20, 20)]

# 맵 설정
walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]

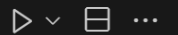
# 게임 루프
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    # 화면 업데이트
    screen.fill((0, 0, 0))
    # 캐릭터 그리기
    screen.blit(character_image, character_position)
    # 적 그리기
    pygame.draw.rect(screen, enemy_color, enemy)
    # 아이템 그리기
    for item in items:
        pygame.draw.rect(screen, (255, 255, 255), item)
    # 맵 그리기
    for wall in walls:
        pygame.draw.rect(screen, (255, 255, 255), wall)
    # 화면 갱신
    pygame.display.flip()
    clock.tick(60)

```



Welcome

prac01.py



C: > Users > sunje > Desktop > Commit > prac01.py > [🔗] items

```
65 # 적 이동
66 enemy.x += enemy_speed[0]
67 if enemy.x < 400:
68     enemy_speed[0] = 1
69 if enemy.x > 600:
70     enemy_speed[0] = -1
71 enemy.y += enemy_speed[1]
72 if enemy.y < 300:
73     enemy_speed[1] = 1
74 if enemy.y > 500:
75     enemy_speed[1] = -1
76
77 # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
78 if character.collidect(enemy):
79     enemy_color = (255, 255, 0)
80     enemy.x -= enemy_speed[0]
81     enemy.y -= enemy_speed[1]
82
83 screen.fill((128, 128, 128)) # 배경색
84
85 for wall in walls:
86     pygame.draw.rect(screen, (128, 0, 0), wall)
87
88 for item in items:
89     pygame.draw.rect(screen, (0, 128, 0), item)
90
91 pygame.draw.rect(screen, enemy_color, enemy)
92 #pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
93 screen.blit(charcter_image, charcter_position)
94 pygame.display.update()
95
96 clock.tick(60) # 60 FPS
97
98 pygame.quit()
```





연습 문제 (9)

- 아이템을 획득하면(아이템과 충돌), 아이템 제거하기



Welcome

prac01.py

prac01-7.py 끝

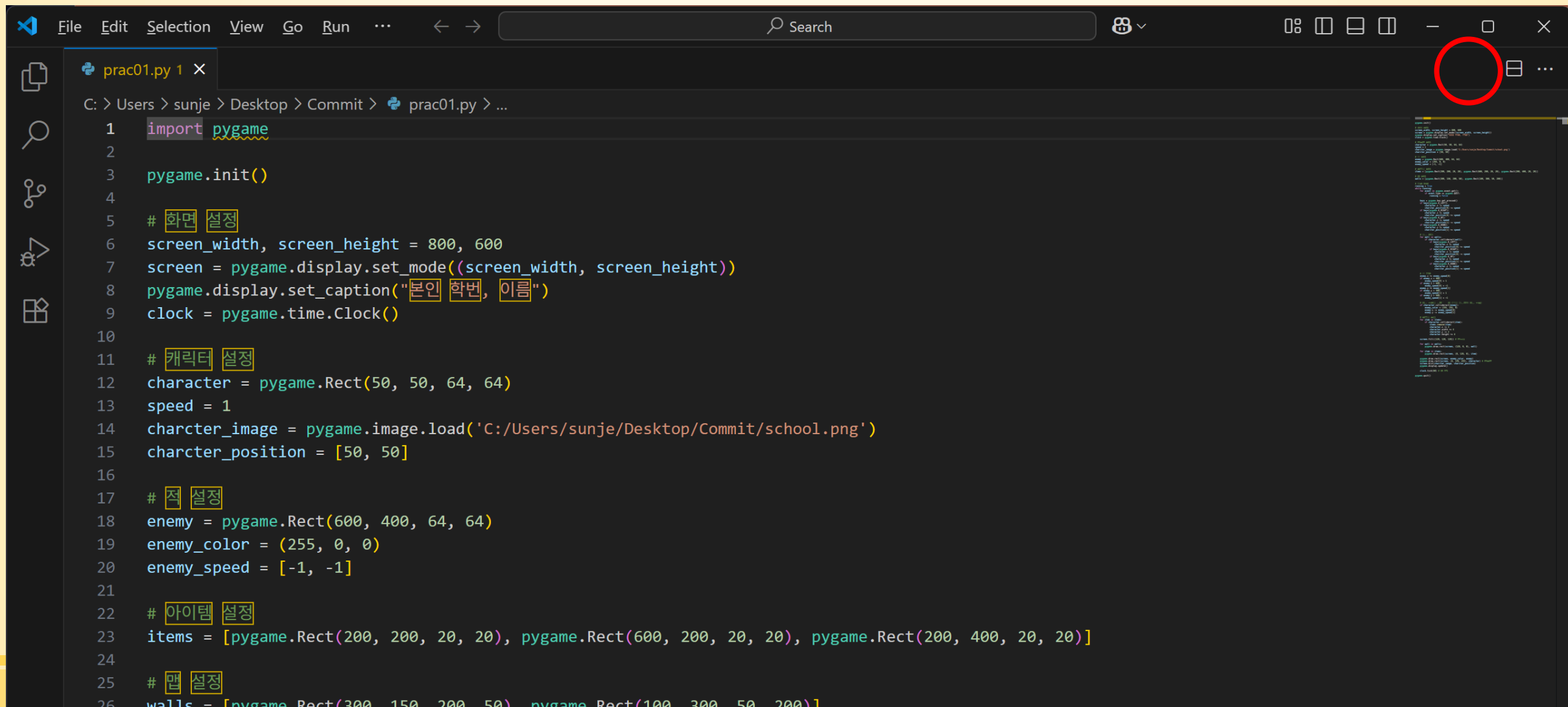
C: > Users > sunje > Desktop > Commit > prac01.py > ...

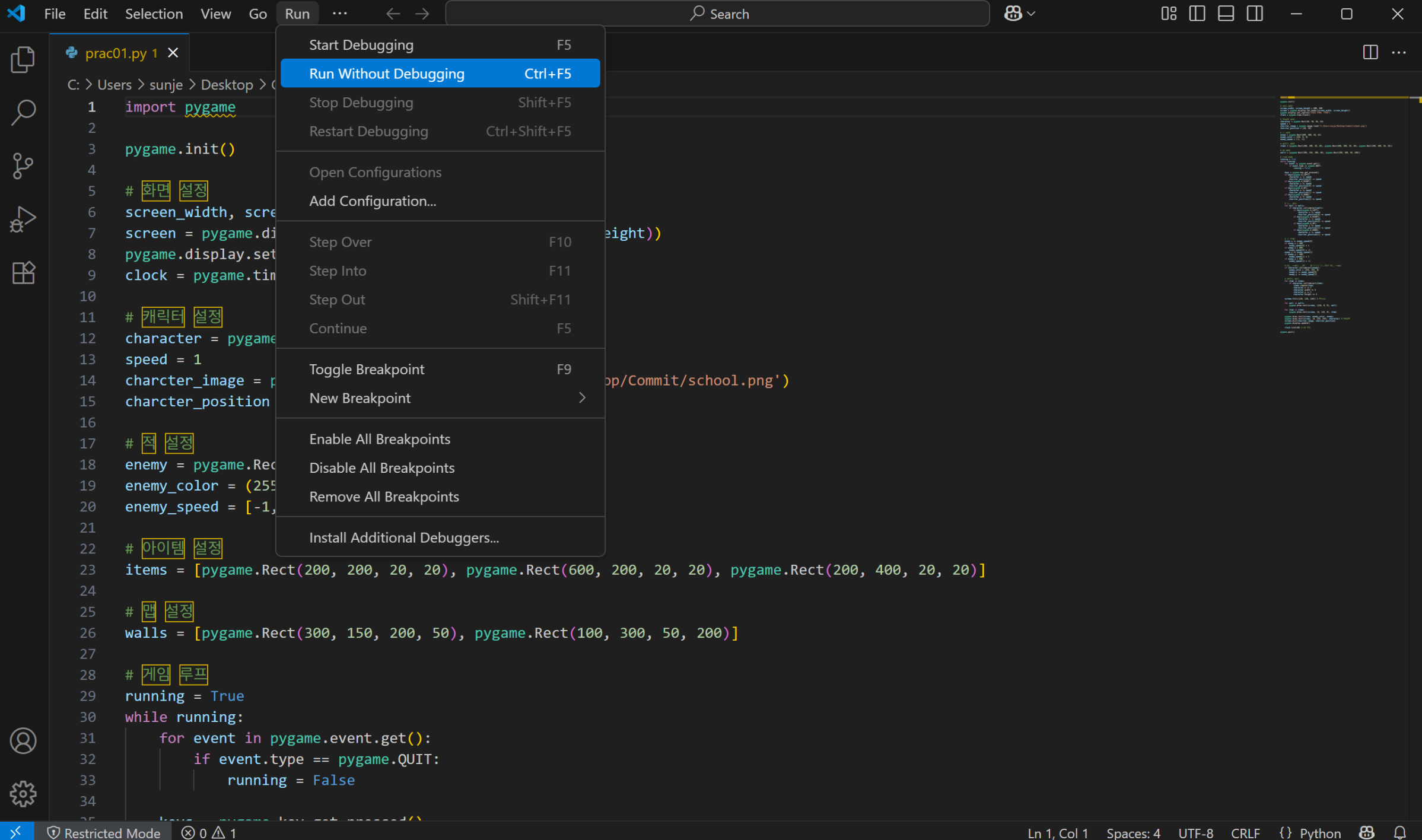
```
74     if enemy.y > 500:
75         enemy_speed[1] = -1
76
77     # 전투 시스템 구현할 부분 (적과 충돌하면 전투 시작)
78     if character.colliderect(enemy):
79         enemy_color = (255, 255, 0)
80         enemy.x -= enemy_speed[0]
81         enemy.y -= enemy_speed[1]
82
83     # 아이템 획득
84     for item in items:
85         if character.colliderect(item):
86             items.remove(item)
87             character.x -= 2
88             character.width += 4
89             character.y -= 2
90             character.height += 4
91
92     screen.fill((128, 128, 128)) # 배경색
93
94     for wall in walls:
95         pygame.draw.rect(screen, (128, 0, 0), wall)
96
97     for item in items:
98         pygame.draw.rect(screen, (0, 128, 0), item)
99
100    pygame.draw.rect(screen, enemy_color, enemy)
101    pygame.draw.rect(screen, (0, 128, 255), character) # 캐릭터
102    screen.blit(charcter_image, charcter_position)
103    pygame.display.update()
104
105    clock.tick(60) # 60 FPS
106
107    pygame.quit()
```

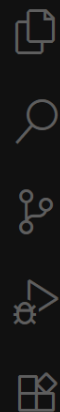
```
108    # 게임 종료
109    # pygame.quit()
110    # sys.exit()
111
112    # 게임 실행
113    while True:
114        for event in pygame.event.get():
115            if event.type == pygame.QUIT:
116                pygame.quit()
117                sys.exit()
118
119            # 키보드 입력
120            if event.type == pygame.KEYDOWN:
121                # 방향키
122                if event.key == pygame.K_LEFT:
123                    # 왼쪽으로 이동
124                    character.x -= 1
125
126                elif event.key == pygame.K_RIGHT:
127                    # 오른쪽으로 이동
128                    character.x += 1
129
130                elif event.key == pygame.K_UP:
131                    # 위로 이동
132                    character.y -= 1
133
134                elif event.key == pygame.K_DOWN:
135                    # 아래로 이동
136                    character.y += 1
137
138                # 스페이스바
139                elif event.key == pygame.K_SPACE:
140                    # 전투 시작
141                    # 전투 시스템 구현할 부분
142
143            # 마우스 입력
144            if event.type == pygame.MOUSEBUTTONDOWN:
145                # 마우스 클릭
146                if event.button == 1:
147                    # 왼쪽 버튼 클릭
148                    # 전투 시작
149                    # 전투 시스템 구현할 부분
150
151        # 게임 실행
152        # pygame.draw.rect(screen, enemy_color, enemy)
153        # pygame.draw.rect(screen, (0, 128, 255), character)
154        # screen.blit(charcter_image, charcter_position)
155        # pygame.display.update()
156
157        # 게임 실행
158        clock.tick(60)
159
160    pygame.quit()
161    sys.exit()
```



(참고) 갑자기 Run ▶ 버튼이 안 보이면?







prac01.py 1 X

C: > Users > sunje > Desktop > Commit > prac01.py > ...

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load("character.png")
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = [-1, -1]
21
22 # 아이템 설정
23 items = [pygame.Rect(200, 200, 20, 20), pygame.Rect(600, 200, 20, 20), pygame.Rect(400, 20, 20, 20)]
24
25 # 맵 설정
26 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
27
28 # 게임 루프
29 running = True
30 while running:
31     for event in pygame.event.get():
32         if event.type == pygame.QUIT:
33             running = False
34
35     pygame.display.flip()
36     clock.tick(60)
```



Do you trust the authors of the files in this workspace?

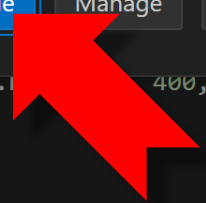
Running executes build tasks and program code from your workspace.

If you don't trust the authors of these files, we do not recommend continuing as the files may be malicious. See [our docs](#) to learn more.

Trust Workspace & Continue

Manage

Cancel





prac01.py ×

C: > Users > sunje > Desktop > Commit > pr

```
1 import pygame
2
3 pygame.init()
4
5 # 화면 설정
6 screen_width, screen_height = 800, 600
7 screen = pygame.display.set_mode((screen_width, screen_height))
8 pygame.display.set_caption("본인 학번, 이름")
9 clock = pygame.time.Clock()
10
11 # 캐릭터 설정
12 character = pygame.Rect(50, 50, 64, 64)
13 speed = 1
14 character_image = pygame.image.load('C:/Users/sunje/Desktop/Commit/school.png')
15 character_position = [50, 50]
16
17 # 적 설정
18 enemy = pygame.Rect(600, 400, 64, 64)
19 enemy_color = (255, 0, 0)
20 enemy_speed = [-1, -1]
21
22 # 아이템 설정
23 items = [pygame.Rect(200, 200, 20, 20), pygame.Rect(600, 200, 20, 20), pygame.Rect(200, 400, 20, 20)]
24
25 # 맵 설정
26 walls = [pygame.Rect(300, 150, 200, 50), pygame.Rect(100, 300, 50, 200)]
27
28 # 게임 루프
29 running = True
30 while running:
31     for event in pygame.event.get():
32         if event.type == pygame.QUIT:
33             running = False
34
35     pygame.key.set_repeat(0)
```

Select debugger

Python Debugger

More Node.js options...

More Python Debugger options...

Install an extension for Python...

